MIL-B-52734B(ME)
5 October 1983
SUPERSEDING
MIL-B-52734A(ME)
15 December 1975

MILITARY SPECIFICATION

BRIDGE, FLOATING: RIBBON, INTERIOR BAY AND RAMP BAY

This specification is approved for use by the USA Belvoir Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers interior bays and ramp bay which, together with the transporter (MIL-T-52735), constitute the ribbon bridge system.

1.2 Classification. The bridge bays shall be of the following types, as specified (see 6.2):

Type I - Interior bay.
Type II - Ramp bay.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5420

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MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-G-20241	- Gasket Material, Wood Felt, Impregnated, Adhesive, Pressure-sensitive.
MIL-P-23377	- Primer Coating, Expoxy Polyamide, Chemical and Solvent Resistant.
MIL-E-23765/2	- Electrodes and Rods - Welding, Bare, Solid, Low Alloy Steel.
MIL-B-43436	- Band, Marker, Blank.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-T-52735	- Transporter: M812 Chassis, For Ribbon Floating Bridge.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-209	- Slinging Eyes and Attachments for Lifting and Tying Down Military Equipment.
MIL-STD-889	- Dissimilar Metals.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

ME

TA13218E4002	- Bridge, Floating: Ribbon, Interior Bay.
TA13218E4005	- Bridge, Floating: Ribbon, Ramp Bay.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which

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are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY, INC. (AWS)

A5.1 - Mild Steel Covered Arc-Welding Electrodes.
A5.5 - Low-Alloy Steel Covered Arc-Welding Electrodes.
A5.10 - Aluminum and Aluminum-Alloy Welding Rods and Bare Electrodes.
A5.18 - Mild Steel Electrodes for Gas Metal Arc Welding.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 N.W. Seventh Street, Miami, FL 33125.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The bridge bays and bridge bay components shall be in accordance with the applicable top assemblies (see 2.1) and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the requirements shown on the drawings is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly and operation of the bridge bays. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative. The dimensions shown on the drawings apply at temperatures from 68° F to 70° F.

3.2 First article.

3.2.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection and test (see section 4 and section 6).

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3.2.2 Hydraulic system components. In addition to the first article type II bridge bays, the following hydraulic components shall be furnished and tested by the contractor prior to the time the first articles are furnished: One each of all hydraulic fittings, hose assemblies, valves, quick-disconnects, couplings and plugs, and hydraulic cylinders. The hydraulic components shall be identical to those furnished on the first article. Tests shall be those specified in section 4 and shall be subject to surveillance and approval by the Government.

3.3 Material. Material shall be as specified herein and as shown on the applicable drawings. The surfaces of materials used shall be free of foreign matter, scale, rust, or corrosion.

3.3.1 Material deterioration and control. Type I and II bridge bays shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a coat of epoxy primer conforming to MIL-P-23377, or corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.3.2 Welding electrodes. Unless otherwise specified on the end product drawings, the following welding electrodes shall be used.

3.3.2.1 For aluminum. Welding electrodes for aluminum shall conform to AWS A5.10, classification ER5356.

3.3.2.2 For steel.

3.3.2.2.1 High-strength steel. Welding electrodes for high-strength steel, including joints with steel castings, shall conform to AWS A5.5, classification E12015, E12016, E12018-M or MIL-E-23765/2, classification MIL-120S-1.

3.3.2.2.2 Mild steel. Welding electrodes for mild steel, including joints with steel castings, shall conform to AWS A5.1, classification E7015, E7018, E7028, or AWS A5.18, classification E70 S-3.

3.4 Hydraulic oil. Oil for all hydraulic systems shall conform to MIL-L-2104, grade 10.

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3.5 Ramp bay hydraulic system. The assembled hydraulic system shall withstand pressures as specified in 3.5.1, 3.5.2 and 3.5.5 without external leakage (except weepage normally expected at piston rod packing), permanent deformation or rupture to any part of the system. Prior to delivery, the hydraulic system shall be filled with hydraulic oil conforming to MIL-L-2104, Grade 10. The hose age shall not exceed 12 quarters from the date of cure to the date of assembly and not more than 8 quarters from the date of assembly to the date of acceptance of the bays by the Government. Separately procured hose assemblies shall be provided with suitable plastic protectors for end openings and threads.

3.5.1 Hydraulic cylinder. When tested in accordance with 4.5.2.4, the hydraulic cylinder shall withstand a pressure of 9700 pounds per square inch (psi) on the piston rod end and a pressure of 1500 psi on the cap end without external leakage, permanent deformation or rupture of any part of the system.

3.5.2 Hydraulic pump. When tested in accordance with 4.5.2.8, the hydraulic pump shall withstand a pressure of 1500 psi without external leakage anywhere in the pressurized system, permanent deformation or rupture of any part within the system. All hydraulic lines and components shall be free of any foreign matter, such as metal chips and filings, that may in any way contaminate the oil or damage the system.

3.5.3 Fittings and hydraulic hose. When tested in accordance with 4.5.2.8, fittings and hydraulic hose shall conform to the individual end product drawing requirements and shall withstand the pressure for the system of which they are a part.

3.5.4 Quick-disconnect couplings. When tested in accordance with 4.5.2.8, quick-disconnect couplings and plugs shall conform to the individual end product drawing requirements and shall withstand the pressure specified for the system of which they are a part.

3.5.5 Control valve. When tested in accordance with 4.5.2.6, all of the working sections of the hydraulic pump control valve shall withstand a pressure of 9700 psi without evidence of valve binding or sticking.

3.5.6 Pressure relief valve. When tested in accordance with 4.5.2.5, the pressure relief valve shall be as specified on the applicable end product drawing. In addition, the valve shall withstand and operate at the applied pressure without malfunction or failure.

3.5.7 Internal leakage. When tested in accordance with 4.5.2.7, the hydraulic oil leakage during the hydraulic cylinder pressure test shall not exceed the rate specified on D13218E4078.

3.6 Threads. All threads not otherwise covered by this specification or the end product drawings shall conform to FED-STD-H28.

3.7 Hydraulic system cleanliness. Prior to installation into the ramp bay, the interior and exterior surfaces of all hydraulic components shall be washed and flushed to remove all visible contaminants, particles and foreign elements.

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Openings of the components shall be immediately covered until installation. The hydraulic system shall be operated and bled prior to being subjected to the tests specified herein. The hydraulic system shall be kept clean of foreign matter and contamination during all tests and when tested in accordance with 4.5.2.12 shall meet the contamination limits specified in table I prior to acceptance.

Table I. Contamination limits (particles per milliliter).

Quantity (max)	Particle Size
1000	Greater than 10 microns
100	Greater than 20 microns

3.8 Performance. When tested in accordance with 4.5.2.11, the bridge bays shall perform as specified herein without evidence of any of the following defects:

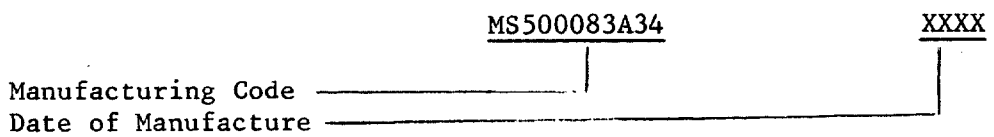
a. Malfunctioning of components or failure of bridge bay to unfold and fold as required.
b. Binding of parts during launching and retrieving.
c. Permanent distortions (other than normal wear).
d. Material defects such as cracks or open joints.
e. Failure of latches and pin-drive mechanism to function as required.
f. External hydraulic leakage.
g. Failure of hydraulic systems to operate as required.
h. Bridge bays not connecting.
i. Water leaking into the pontons.
j. Ramp bays not capable of being articulated to 18 degrees from the horizontal.

3.9 Lubrication. Lubrication shall be as shown on the end product drawings or as specified herein.

3.10 Cleaning, treatment and painting. Cleaning, treatment and painting shall be as shown on the end product drawings. Where MIL-T-704, Type B paint is required, the finish coat shall be camouflage green 383 conforming to MIL-C-46168.

3.11 Identification marking. The bridge bays and bridge bay components shall be identified in accordance with MIL-STD-130 and as indicated on the end product drawings.

3.11.1 Identification marking for hose assemblies. Each hose assembly shall have identification marking stamped on an aluminum band conforming to MIL-B-43436, except that the band shall be of sufficient width to allow a minimum of two lines of 1/8-inch metal stamping. Unless otherwise specified on the drawings, the marking band shall contain the following information:



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3.12 Stenciling. The nomenclature, National Stock Number, weight, and cubage shall be stenciled on the end of each ponton. The letters "NET WT" shall precede the numerals in the weight and the letters "CU" shall precede the cubic displacement. Markings shall be Gothic-type capitals and Arabic numerals 3/4 inch high. Stenciling shall conform to MIL-T-704. Weight of ramp roadway pontons shall include hydraulic system and approach ramp.

3.13 Government-loaned property. Unless otherwise specified (see 6.2), the following property, in the quantities indicated, will be loaned by the Government (see 6.6):

Item No.	Quantity	Description	Identification
1	1	Transporter, Ribbon Bridge	MIL-T-52735
2	1	Test Fixture, Hydraulic Cylinder	13218E4319
3	1	Frame, Pull Test	13219E4302
4	1	Ramp Bay Hydraulic Cylinder 1/	13218E4078
5	as required	Bridge Bay, type I or II 2/	13218E4002 13218E4005
6	1	Quality Assurance Pamphlet, "Inspection Equipment for Bridge, Floating: Ribbon, Interior Bay and Ramp Bay" 3/	P715-141 (included in TA's 2.1)
7	1	Technical Manual, Operator's and Organizational Maintenance, "Improved Float Bridge (Ribbon Bridge)".	TM-5420-209-12

1/ When type I bays are procured separately, the Government will furnish a cylinder for use with pull test.
2/ When type I or II bays are procured separately, the Government will loan a sufficient number of other types of bays to accomplish tests.
3/ The contractor is responsible for fabricating the functional inspection equipment and obtaining the necessary optical inspection equipment required by Quality Assurance Pamphlet P715-141.

3.14 Ponton leak. When tested in accordance with 4.5.2.10 each individual ponton shall be capable of maintaining a constant, independent, internally applied air pressure of 1.5 psi, plus or minus 0.1 psi for at least 5 minutes without leakage.

3.15 Pull. When tested in accordance with 4.5.2.9, each interior roadway ponton shall be capable of withstanding a tensile load of 483,000 pounds for a time period of 5 minutes without evidence of permanent deformation or structural failure anywhere within the ponton.

3.16 Reliability. When tested in accordance with the test plan appearing in 4.5.2.13 an individual type I or type II bay shall possess the following mission specified reliabilities (expressed as percent probability of success):

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a. Type I bridge bay (interior bay), 99.75 percent probability of successfully completing a mission (see 4.5.2.13.1). A mission requiring 12 type I bays results in a specified mission reliability of 97.0 percent when the 12 type I bays are considered as a subsystem (see test plan I, III, or IV, as applicable).
b. Type II bridge bay (ramp bay), 98.0 percent probability of successfully completing a mission (see 4.5.2.13.1). A mission requiring two type II bays results in a specified mission reliability of 96.0 percent when the two type II bays are considered as a subsystem (see test plan II, III, or IV, as applicable).

3.17 Maintainability. The ratio of the manhours of maintenance required to the hours of operation performed shall not exceed .09 and .13 for the interior bay and ramp bay, respectively, when tested per 4.5.2.13. The man-hours of maintenance shall include both preventive and repair maintenance manhours.

3.18 Slinging and tiedown provisions. When tested in accordance with 4.5.2.14, slinging and tiedown attachments shall be as shown on the drawings and shall meet the requirements of MIL-STD-209.

3.19 Interchangeability. All parts, components and assemblies having the same identification number shall be functionally and dimensionally interchangeable. Interchangeability is defined as two or more parts (components and assemblies) possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of themselves or of adjoining parts (components and assemblies), except for built-in adjustment, and without selection for fit or performance.

3.20 Specification-control items. Prior to using any specification-control item other than a suggested source item, the contractor shall furnish to the contracting officer evidence that the item conforms to all requirements on the applicable drawing. The contractor is responsible for performing all inspections and tests on specification or source control items if not performed by the vendor. No change shall be made to any source or specification control item after approval of the preproduction model without prior approval by the contracting officer.

3.21 Type I, interior bay. The type I bays shall conform to top assembly, TA13218E4002.

3.22 Type II, Ramp Bay. The type II bays shall conform to top assembly, TA 13218E4005.

3.23 Workmanship. All members, components and assemblies containing moving parts shall be assembled free of harmful extraneous materials such as metallic particles, dirt, grit, or corrosion.

3.23.1 Castings. All casting used on the bridge bays shall be cleaned of harmful extraneous material, such as sand, dirt, pits, scale, and flux. Rework shall be limited to procedures which do not reduce strength or affect function.

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3.23.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing for steel only. Aluminum alloy shall not be cut by methods using heat or heated for forming. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. All parts shall be completely and accurately shaped to the drawing requirements prior to joining (welding). Joints, butts and splices shall be limited to those shown on the end product drawings. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.23.3 Welding. Unless otherwise specified on the end product drawings, welding shall be done in accordance with the requirements of this specification. Prior to the fabrication of any weldment, the contractor shall prepare and submit to the contracting officer for approval, an isometric or perspective drawing showing the location of each joint and the joint welding procedures to cover all welding, including a general outline for the repair of base material and welded joints. These procedures shall also address the proposed method of surface preparation prior to welding. The procedures shall be qualified in accordance with the ASME Boiler and Pressure Vessel Code, Section IX, except that the guided bend test is not required for steel castings. All welds to be qualified shall be made on the same alloy, gage and temper of material, with the same filler materials, and in the same position as will be used in production. Weld procedures which have been qualified under other equivalent recognized codes may be considered qualified for the purposes of this specification, provided they are approved by the contracting officer. Any change in procedure during fabrication shall require requalification in accordance with the requirements of the ASME Boiler and Pressure Vessel Code, Section IX.

3.23.3.1 Aluminum welding. The aluminum surfaces of parts to be welded shall be free from scale, paint, grease, oxide film and other foreign matter. The welding shall be done by the gas-metal-arc (MIG) method, except as shown on end product drawings. Where material gage or size prevents use of MIG welding, the gas-tungsten-arc (TIG) method may be employed. Welds may be ground, filed, buffed, or chipped, but not hammered, and shall be free from defects, such as base metal cracks adjacent to or behind welds, cracks in weld metal, undercutting of base metal, lack of fusion with parent metal or between parts, porosity, lack of penetration, lack of prescribed fit, inclusions, or spatter. To assure that all welding variables have been addressed, the welding procedures required by 3.23.3 shall include provisions for a daily on-the-job aluminum weldbreak test by each production welder during each work shift. The procedure shall require that the welder construct the "weldbreak" specimen using the same materials, alloys, and thicknesses, with the same weld type and size, and cleaned, prepared and welded in accordance with the applicable procedures and techniques developed for the end item weldment being welded at the time the break test is required. The procedures shall also include the steps to be followed for breaking the specimens and examination of the fractured parts to determine the weld quality. The

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procedures shall also describe the corrective action to be taken in the event that the specimen test demonstrates weld quality that is not acceptable, including disposition of component welding accomplished by the welder since the last acceptable weldbreak test.

3.23.3.2 Steel welding. The surfaces of steel parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the bridge bays to conform to the examination and tests requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch unless appropriate bridging techniques are used.

3.23.3.3 Care of welding electrodes. Care of welding electrodes specified for welding high-strength steel shall be as follows:

a. When a container of welding electrodes is opened, the entire contents shall be placed immediately in a baking oven, with electrodes placed not more than three layers deep on a tray, and baked either (1) for a minimum of 1 hour, but not more than 1-1/4 hours, at 800° F; or (2) for a minimum of 4 hours at 400° F.
b. The baked electrodes, while still warm, shall then be placed in a ventilated holding oven set at 250° F to 300° F.
c. Not more than a 1/2-hour supply of welding electrodes shall be taken from the holding oven at one time.
d. If electrodes are out of the holding oven in excess of 1/2 hour but less than 1 hour, they shall be returned to the holding oven for a period of not less than 1 hour.
e. If electrodes are out of the holding oven in excess of 1 hour, they shall be returned to the baking oven and rebaked as described under a.
f. Electrodes removed for use shall not be allowed to come in contact with any moist or wet surfaces. Electrodes that have been in contact with water, grease or oil shall not be used.

3.23.4 Welders. Before assigning any welder to fabricate bridge bays covered by this specification, the supplier shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by the ASME Boiler and Pressure Vessel Code, Section IX, for the type of welding operations to be performed and that such qualification is effective as defined by the code. Test specimens shall be made on the same alloy, gage and temper of material with the same filler materials and in the same position as will be used in production. Welders who have been similarly qualified under other recognized codes may be considered qualified for the purposes of this specification, provided they are approved by the contracting officer.

3.23.5 Machine work. Tolerances and gages for metal fits shall conform to the limits specified on the applicable end product drawings.

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3.23.6 Jigs and fixtures. Fabrication, assembly and installation of all bridge bay components (except minor parts) shall be accomplished with production tooling, jigs or fixtures, and parts shall be joined while held in position. Jigs or fixtures shall be designed to minimize distortion of the parts being welded and to assure that the configuration, alinement and location of parts meet the specified requirements. Subsequent straightening and thermal stress relieving is permitted to relieve distortion, provided the straightening process does not damage the component.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with referenced specifications, standards and drawings, as applicable.

4.1.2 Disassembly inspection. Failure of any examination or test by the first article shall be cause for disassembly, in the presence of a Government representative, of the first article to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first article. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the quality assurance provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring QAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.7 and 6.8).

4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article bridge bays or separately procured components shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

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4.3.2 Tests.

4.3.2.1 Hydraulic system components. Prior to installation of the hydraulic system components into the first article type II bridge bay, the components shall be tested as specified in 4.5.2.1 through 4.5.2.8. Failure of any test shall be cause for performing the disassembly inspection specified in 4.1.2.

4.3.2.2 Assembled bridge bay. After successful completion of the tests specified in 4.3.2.1, the hydraulic system components shall be installed in the first article type II bridge bay, and the bridge bay shall be tested as specified in 4.5.2.10 through 4.5.2.12. The assembled type I bridge bay shall be tested as specified in 4.5.2.9 through 4.5.2.11. Failure of any test shall be cause for performing the disassembly inspection specified in 4.1.2.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests for separately procured bridge bay components shall be in accordance with MIL-STD-105.

4.4.2 Examination.

4.4.2.1 Individual. Each bridge bay shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2.2 Samples. Samples of separately procured bridge bay components selected in accordance with 4.4.1 shall be examined for the applicable defects specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 6.5 percent defective for minor defects.

4.4.3 Tests.

4.4.3.1 Individual. Each type II bridge bay shall be tested as specified in 4.5.2.1 through 4.5.2.8 and 4.5.2.10 through 4.5.2.12. Each type I bridge bay shall be tested as specified in 4.5.2.9 through 4.5.2.11. Failure of any test shall be cause for rejection.

4.4.3.2 Samples. Samples of separately procured components selected in accordance with 4.4.1 shall be subjected to the applicable test specified in 4.5.2.1 through 4.5.2.12. AQL shall be 2.5 percent defective.

4.5 Inspection schedule.

4.5.1 Examination. Each type of bridge bay or separately procured components as specified shall be examined as specified herein for the following defects as applicable:

Major

101. Any part or component not in accordance with QAP requirements as shown on the end product drawings.
102. Any dimension other than those identified on the QAP requirements shown on the drawings not as specified.

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103. Incorrect bolt torque.
104. Members not true to dimension and line or containing twists, bends or other imperfections.
105. Weld defects such as base metal cracks adjacent to or behind welds, cracks in weld metal, undercutting of base metal, lack of fusion with parent metal or between parts, porosity, lack of penetration, lack of prescribed fit, incorrect preparation, slag inclusion, spatter, or incorrect size.
106. Qualification of welders or weld procedure not as specified.
107. Welding not performed in accordance with qualified procedure.
108. Material not as specified.

a. Materials are not corrosion resistant or treated to be made corrosion resistant for the applicable storage and operating environment.
b. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
c. Contractor does not have documentation available for identification of material, material finishes or treatment.

109. Hydraulic system cleanliness not as specified.
110. Commercial components not as specified.
111. Components not interchangeable with respect to installation, performance and quality.
112. Nonskid compound not applied correctly or missing.
113. Leaks in hydraulic lines.
114. Quick-disconnect couplings not as specified.
115. Age of hose not as specified.
116. Components missing or not as specified.
117. Workmanship not as specified.

Minor

201. Treatment and painting not as specified.
202. Lubrication and lubrication fittings not as specified.
203. Identification marking and hose assembly identification missing or incorrect.
204. Warning and instruction signs and decals missing.

4.5.2 Tests.

4.5.2.1 Test conditions. All tests, with the exception of the operational performance test and the reliability test, shall be conducted at an ambient air temperature of from plus 25° F to plus 100° F with the hydraulic oil temperature between plus 25° F and plus 90° F. The operational performance and reliability tests may be conducted at existing ambient temperatures, except that the contractor shall be responsible for insuring that icing conditions do not impair performance of the operational test. Hydraulic oil conforming to MIL-L-2104, Grade 10, shall be used for all tests. Prior to assembly of hydraulic components for test or installation into bridge bays, all hydraulic lines and hydraulic

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components shall be free of any foreign matter, such as metal chips and filings, that may in any way contaminate the oil or damage the system. All equipment shall be operated in accordance with the applicable technical manual.

4.5.2.2 Test schedule. All tests shall be conducted in accordance with table II and in the specified sequence.

TABLE II. Test Schedule

First Article	Quality Conformance	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5
X	X	Test conditions	4.5.2.1	
X	X	1/ Limbering up of hydraulic cylinder	4.5.2.3	
X	X	1/ Hydraulic cylinder pressure	4.5.2.4	3.5.1
X	X	1/ Cylinder relief valve	4.5.2.5	3.5.6
X	X	1/ Control valve	4.5.2.6	3.5.5
X	X	1/ Internal leakage	4.5.2.7	3.5.7
X	X	1/ Ramp articulation system pressure	4.5.2.8	3.5.2, 3.5.3, 3.5.4
X	X	2/ Pull	4.5.2.9	3.15
X	X	Ponton leak	4.5.2.10	3.14
X	X	Operational performance	4.5.2.11	3.8
X	X	1/ Post-test hydraulic cleanliness	4.5.2.12	3.7
-	-	Reliability and maintainability	4.5.2.13	3.16 and 3.17
-	-	Sling and tiedown provisions	4.5.2.14	3.18

1/ Applies to type II bridge bays only.
2/ Applies to type I bridge bays only.

4.5.2.3 Limbering up of hydraulic cylinder. Prior to subjecting the cylinder to the tests specified herein, the cylinder shall be cycled through 25 complete cycles. One cycle shall consist of extension and retraction of the rod through full travel without bottoming. The cycling shall be done by either hydraulic pressure or mechanical means.

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4.5.2.4 Hydraulic cylinder pressure. Place the cylinder in the test frame (see 3.13) so that the piston is restrained from bearing against either cylinder head. In accordance with 13218E4078, note 5, subject Port "A" of the cylinder to 9700 psi, plus 0 psi minus 200 psi, for 5 minutes. Nonconformance to 3.5.1 shall constitute failure of this test.

4.5.2.5 Cylinder relief valve. The hydraulic cylinder shall be pressurized until the relief valve opens. Do not pressurize the cylinder above 9700 psi. Opening of the valve at any pressure below 9500 psi or failure to open before 9700 psi is reached shall constitute failure of this test. After passing the above test, the pressure shall be reduced to 7000 psi. Failure of the valve to close (as indicated by an internal leakage of no more than 50 drops a minute at Port "B") before 7000 psi is reached or nonconformance to 3.5.6 shall constitute failure of this test.

4.5.2.6 Control valve. The hydraulic pump control valve shall be subjected to a pressure between 9700 psi and 9900 psi. Evidence of leakage or permanent deformation of the valve or nonconformance to 3.5.5 shall constitute failure of this test.

4.5.2.7 Internal leakage. During the hydraulic cylinder pressure test, leakage shall be measured at Ports "A" and "B", as specified on Drawing D13218E4078 of TA218E4005. Leakage in excess of the rate specified on D13218E4078 shall constitute failure of this test.

4.5.2.8 Ramp articulation system pressure. Subject the hydraulic pump, hose and hydraulic cylinder (Ports "A" and "B") assembly to a pressure between 1500 psi and 1600 psi for least 1 minute. Evidence of external leakage, permanent deformation or nonconformance to 3.5.2, 3.5.3 or 3.5.4 shall constitute failure of this test.

4.5.2.9 Pull. Install the interior roadway ponton in the test frame (see 3.13) and apply a tensile load at each end through the ponton eye and yoke (see top assembly, TA13218E4002, drawing D13218E4003 finds 16 and 17). Apply the tensile load through a ramp hydraulic cylinder (see 3.13) pressurized in increments of approximately 925 psi to a pressure of 9250 psi. Hold the maximum load for a period of 5 minutes. Nonconformance to 3.15 shall constitute failure of this test.

4.5.2.10 Ponton leak. Internally pressurize with air each individual ponton of the bridge bay to 1.5 psi, plus or minus 0.1 psi, and maintain pressure while soap testing for leaks. If leakage occurs, remove the pressure from the ponton, repair leak area in accordance with the instructions on applicable ponton drawings, and repeat the leak test for each repaired ponton. Evidence of any leakage that cannot be repaired shall constitute failure of this test. After all identified leaks have been repaired, the ponton is to be pressurized to 1.5 psi, plus or minus 0.1 psi, and the air source is to be disconnected and the pressure held for five minutes. Nonconformance to 3.14 shall constitute failure of this test.

4.5.2.11 Operational performance. When ramp bays and interior bays are procured together tests shall be performed on both types of bay. With the

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bridge bay in folded position, adjust unfolding cables to tension specified on the assembly drawings. Using the ribbon bridge transporter (see 3.13), free-launch the bridge bay in a body of water of sufficient size and depth (a minimum of 4 feet) to accommodate a free-launch without damage to the bridge bay. With the bridge bay unfolded and floating, engage the roadway-to-roadway deck connectors and the bridge latches. For the ramp bay, the roadway pontons will have to be pulled together for these connectors to engage their mating receptacles. The bridge bay shall remain floating for a minimum time period of 30 minutes, after which the pontons shall be inspected for leakage. While the bridge bay is floating, operate the pin-drive mechanism to insure free movement for the entire pin travel in both directions. The operational performance test shall include connecting both ends of the interior bay to the ramp bay. When interior bays are procured separately, two first article bays shall be provided for the connecting test. When ramp bays are procured separately, the Government shall furnish an interior bay for the connecting test (see 3.13). Prior to retrieval, each ramp bay yoke will be articulated to a minimum of 18 degrees from the vertical and will then be pumped back to the vertical position. Prepare the bridge bay for retrieval and retrieve the bridge bay onto the transporter. Failure of this test shall consist of any of the following defects:

a. Failure of roadway-to-bow foldlock latches to retain bows in folded position while on transporter.
b. Failure of roadway-to-roadway foldlock latches to retain the roadway pontons in folded position while on transporter.
c. Failure of launched bridge bay to unfold due to action of unfolding mechanism without any other aids.
d. Failure of the roadway-to-bow bridge latches to engage their mating bow receptacles when operated manually.
e. Evidence of ponton leakage.
f. Failure of connectors or pin-drive mechanism to function as intended.
g. Failure of bridge bay to fold during retrieval without any assistance other than the winch cable, or binding during folding.
h. Failure of all foldlock latches to engage completely and automatically when the bridge bay is retrieved.
i. Failure of the ramp yokes to be pumped as intended.
j. Failure of the bays to connect as intended.

Failure of latches to function shall be remedied by adjustment where possible.

4.5.2.12 Post-test hydraulic cleanliness. After completion of the operational performance test and prior to delivery of the type II bridge bays to the Government, a sample of hydraulic fluid shall be taken from one of the quick-disconnect couplings and tested for the particle count specified in table I. If the particle count exceeds the values in table I, the system shall be flushed until contamination is reduced to the acceptable level. Inability to conform to the cleanliness levels specified in table I shall constitute failure of this test.

4.5.2.13 Reliability/maintainability. Using the probability of mission success specified in 3.16, the bay or bays shall be tested as specified herein. Using the maintenance ratios of man-hours of maintenance to hours of operation

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performed for the interior and ramp bay respectively specified in 3.17, the maintainability shall be determined as specified herein. This test will be conducted by the Government at a site selected by the Government. The test bridge system shall consist of a minimum of 12 interior bays, two ramp bays, and 14 transporters (see 3.13). Testing may be performed in still water or in water currents up to 8 feet per second. Maintenance shall be performed in accordance with the technical manual, and at the end of each day's mission, any damage to the bridge bays shall be repaired prior to the next day's mission. Failure to meet all performance requirements of this specification, when such failure is due to faulty material or workmanship, shall be cause for rejection and shall be cause for refusal by the Government to continue acceptance of production units until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on all units previously produced. Any such material or workmanship deficiencies will be considered prima facie evidence that all units accepted prior to the completion of this test are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5.2.13.1 Reliability-maintainability test cycle. Reliability-maintainability testing shall be conducted in cycles equivalent to 10 normal bridge missions. Each such cycle shall consist of the following:

a. 9 launch/retrieve cycles.
b. 1 launch and bridge assembly.
c. 500 class 60 vehicle crossings.
d. 1 bridge disassembly and retrieval.
e. 200 miles of transporting.

4.5.2.13.2 Reliability. The test plan utilized will depend on the method of procurement (see 6.2). Failure criteria based on 14 bay bridge.

a. Test Plan I - Utilized for interior bays if procured separately. Test for five test cycles (50 missions) as specified in 4.5.2.13.1. If less than two failures occur, reliability is acceptable. If two or three failures occur, continue testing until 10 test cycles are complete. If no more than three failures occur, reliability is acceptable. If any time during the test four failures occur, reliability is not acceptable and the test is terminated.
b. Test Plan II - Utilized for ramp bays if procured separately. Test for five test cycles (50 missions) as specified in 4.5.2.13.1. If less than two failures occur, reliability is acceptable. If from two to four failures occur, continue testing until 10 test cycles are complete. If no more than four failures occur, reliability is acceptable. If at any time during the test five failures occur, reliability is not acceptable and the test is terminated.
c. Test Plan III - Utilized for ramp and interior bays if procured together. Test for five test cycles (50 missions) as specified in 4.5.2.13.1. The acceptance criteria in table III shall be used to determine if reliability is acceptable, if testing shall continue until 10 cycles are complete, or if reliability is not acceptable and the

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test is terminated. If testing is continued to 10 cycles and no more than three interior bay failures and no more than four ramp bay failures occur reliability is acceptable. If at any time during the test, four interior bay failures or five ramp bay failures occur, reliability is not acceptable and the test is terminated.

d. Test Plan IV - Utilized when interior bays, ramp bays and transporters are procured together. Test for five test cycles (50 missions) as specified in 4.5.2.13.1 and evaluate reliability. If required, continue testing to 10 cycles and evaluate reliability again. Failure criteria for bridge bays will be as specified in 4.4.2.13.2c of this specification. Failure criteria for transporters will be as specified in MIL-T-52735.

When two or more end items are procured together, only end items which are in a continue test status after 5 cycles will be evaluated further. Acceptable items may be used to enable test of other items but they will not be re-evaluated for reliability. For example: If after 5 cycles there are 1 ramp failure, 2 interior bay failures and no transporter failures, the ramp and transporter have passed the test. Further testing will be conducted to evaluate the interior bay. All three end items are required to test the interior bay, but the ramp and transporter will not be evaluated further.

Table III. Test plan III acceptance criteria.

	5 CYCLES INTERIOR BAY FAILURES		
	less than 2	2 or 3	4 or more
less than 2	A	C	F
2 to 4	C	C	C
5 or more	F	F	F

A = Accept C = Continue Testing F = Fail

For the purpose of reliability computations a mission failure is defined as follows:

a. Any malfunction of interior or ramp bay which causes bridge traffic to be halted for more than 15 minutes.

b. Any malfunction of an interior or ramp bay which would cause inability to complete a mission, as defined in 4.5.2.13.1, in less than 14 hours (utilization of repair resources available to the parent deployed unit allowable).

c. Any impending interior or ramp bay malfunction which would require maintenance that cannot be deferred until the end of the mission because of the creation of a safety hazard and results in the situation

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described in a or b above. simultaneous related malfunctions are considered one failure. Malfunctions which do not affect mission performance are not considered mission failures.

4.5.2.13.3 Maintainability. The maintenance manhours required to perform all corrective and preventive maintenance occurring during the reliability test (see 4.5.2.13.1) shall be recorded and classified as to whether it was scheduled or unscheduled. The MR requirement(s) of 3.19 shall be assessed using the following formula:

$$MR = \frac{\text{Total scheduled maint manhrs + total unscheduled maint manhrs}}{\text{Total operating time}}$$

Operator/crew checks and services are excluded from this ratio.

4.5.2.14 Slinging and tiedown provisions. With the bay(s) anchored by means other than the tiedown being tested, apply the static loads specified in MIL-STD-209 to each provision. Weld failure, permanent deformation of the bay or its tiedown provisions, or breakage of the provisions shall constitute failure of this test.

4.6 Inspection comparison. The Government may select bridge bays at any time during the contract production period and subject these bridge bays to the examination specified in 4.5.1 and the tests specified in 4.5.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on bridge bays selected at random from those which have been accepted by the Government and will not include the previously inspected first article bridge bays. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.6.1 Inspection failure. Failure of an inspection comparison bridge bay to meet any requirement specified herein during and as a result of the examination and tests specified in 4.6 shall be cause for rejection of the inspection comparison bridge bay and shall be cause for refusal by the Government to continue acceptance of production bridge bays until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on bridge bays previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all bridge bays accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

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4.7 Inspection of packaging. The preservation shall be examined to determine compliance with the requirements of section 5 and the quality conformance requirements of MIL-P-116 and MIL-STD-129 of this specification. The marking shall be examined to determine if any marking is missing, illegible, incorrect, or incomplete in accordance with the referenced document. Any defect shall be cause for rejection.

5. PACKAGING

5.1 Preservation and packing (see 6.9).

5.1.1 Bridge bays (interior and ramp). Each bridge bay shall be folded and secured in the transport position. Prior to folding, the following shall be accomplished.

5.1.1.1 Unfinished surfaces. Unpainted or unfinished exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative.

5.1.1.2 Machined surfaces. Exposed machined metal surfaces for which a grease coating has not been specified on the drawings shall be coated with type P-6 or type P-11 preservative.

5.1.2 Preservatives. Preservatives specified herein shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.1.3 Other components. Other components, such as ramp connecting tool and pump handle, shall be placed and secured as provided for.

5.2 Marking (see 6.9). Marking for shipment and storage shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The bridge bays are intended to be used with transporters conforming to MIL-T-52735. The ramp bays and interior bays can be assembled to form bridges of variable length or rafts. The assembled bridge is capable of carrying class 60 vehicle loads in currents up to 8 feet per second, and the rafts are capable of carrying a class 60 vehicle load in currents up to 9 feet per second.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type of bridge bay required (see 1.2).
c. Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1).
d. When first article bridge bays or separately procured components are required and the number of each type of bridge bay or components required (see 3.2).

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e. When Government-loaned property is not required (see 3.14).
f. Which reliability/maintainability test plan will be utilized (see 4.5.2.13.2).

6.3 First article. When a first article inspection is required, the contractor shall furnish one or more of each type of bridge bay or separately procured components for evaluation and test within a specific time frame. The items to be tested shall be either preproduction models or first production items. Examination and test shall be as specified in section 4. The contracting officer shall include specific instructions in the acquisition documents regarding arrangements for examinations specifying which tests will be conducted by the contractor, which tests if any will be conducted by the government, the time frame for the testing and submittal of the first article documentation, how the first article will be monitored by the Government, and how the first article documents will be submitted and approved. Any changes or deviations of production bridge pays from the approved first article model during production will be subject to the approval of the contracting officer. Approval of the first article does not relieve the contractor of his obligation to furnish bridge bays conforming to this specification.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operational manual to be furnished with each bridge bay.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of bridge bays. A suggested paragraph is as follows:

"Shipment of bridge bays shall include repair parts, maintenance tools, operational instructions, and accessories, unless exceptions are provided elsewhere in the contract."

6.6 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.14.

6.7 Quality assurance provisions (QAP). The contracting officer should require the contractors to maintain records of all QAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirements, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

6.8 Definitions.

6.8.1 Quality Assurance Provisions (QAP). A QAP is a contractual requirement that supplements section 4 of this specification. QAP indicates the

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minimum requirements which must be inspected on the end product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.9 Preservation and packing. Levels of preservation and packing have not been included as the preservation specified is the minimum acceptable for any level of protection, and no packing is required. For purposes of preservation/packing level marking, the bridge bays shall be marked A/A.

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